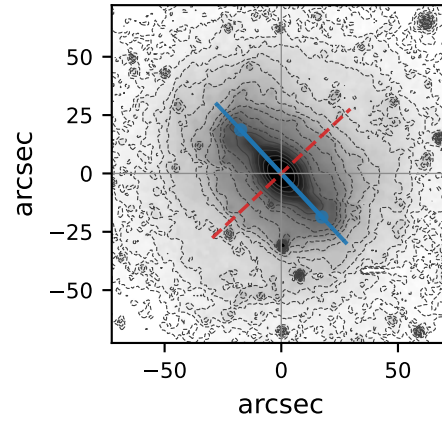
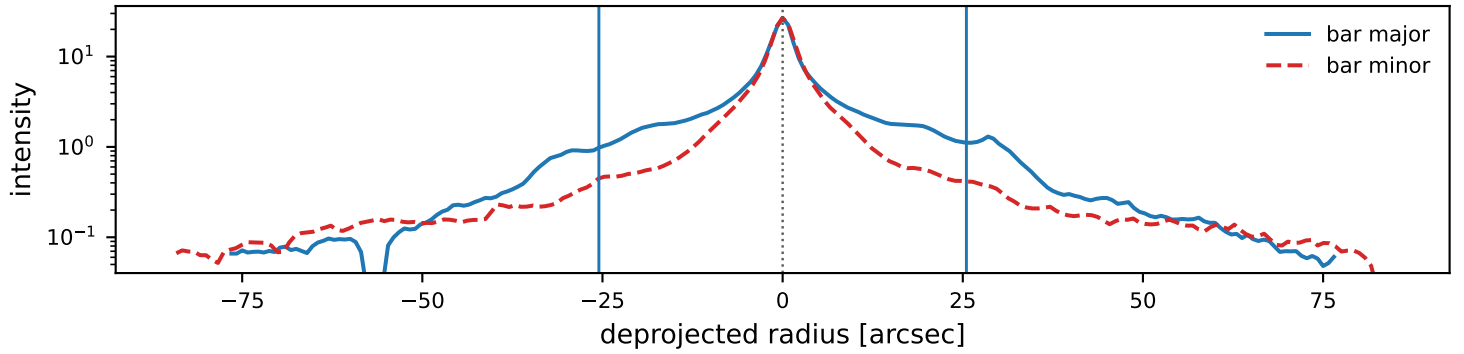


NGC1640 foreground-mask profile comparison bar PA=43.0 deg

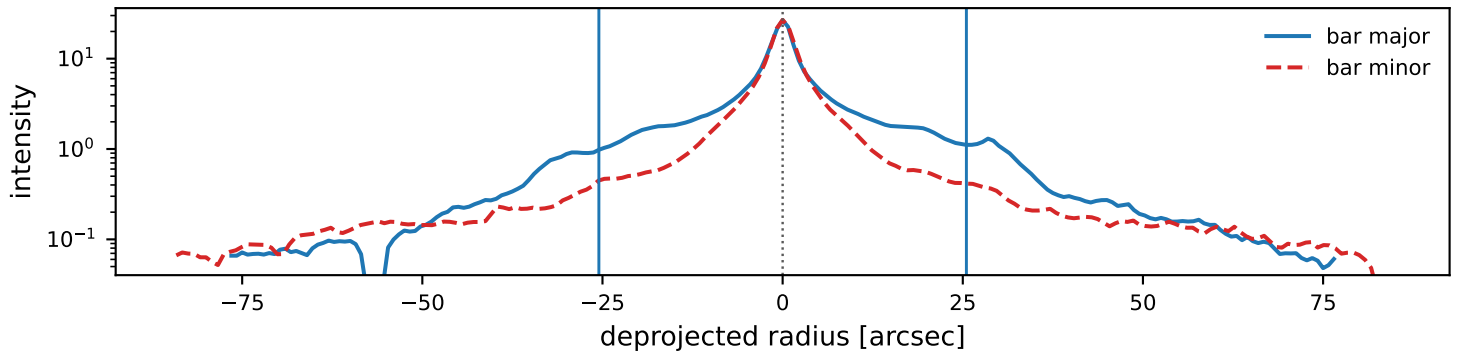
S4G 3.6 micron isophotes with masked objects circled



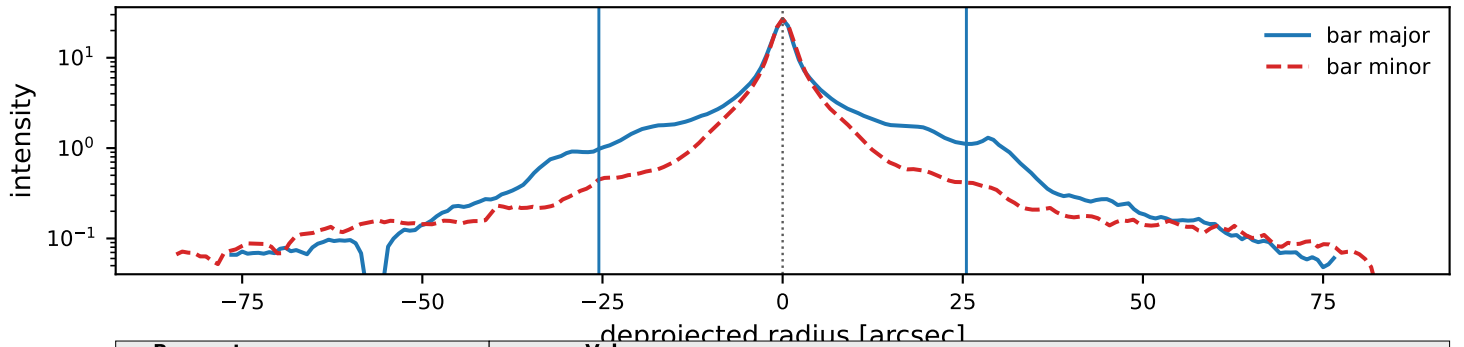
Original major/minor-axis cuts



Masked major/minor-axis cuts



Masked cuts with straight log-linear bridges



Parameter	Value
Masking model	spike-gated; photutils segmentation on residual image; photutils 3.0.0
Residual image	science image - Gaussian-smoothed galaxy model
Detection threshold	5 sigma above residual median
Profile spike gate	0 bar-major spike samples
Spike rule	peak $\geq 25\%$ above 4-15 arcsec neighbours; $\geq 40\%$ above ± 3 samples; $ r \geq 8$ arcsec
Smooth sigma	15 px
Connected-pixel minimum	8 px
Dilation radius	3 px
Max segment area	500 px
Max elongation	6
Central exclusion radius	12 px
Profile width	3 px
Applied mask	0 source segments; 0 pixels ignored
Filled-profile panel	solid=measured data; fine dotted=samples filled by straight log-intensity bridge
Bridge merge gap	12 profile samples
Auto-tune grid	5 sigma; selected 5 sigma